# MAT-MEK4270 Mandatory Assignment 1

Eva Rønningsbakk

## 1.2.3 Exact solution

We consider

$$u(t,x,y) = e^{i(k_x x + k_y y - \omega t)}$$

and show that it satisfies the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right).$$

Differentiating twice with respect to time gives

$$\frac{\partial u}{\partial t} = -i\omega u, \qquad \frac{\partial^2 u}{\partial t^2} = (-i\omega)^2 u = -\omega^2 u.$$

The spatial derivatives are

$$\frac{\partial u}{\partial x} = ik_x u, \qquad \frac{\partial^2 u}{\partial x^2} = -k_x^2 u,$$

and

$$\frac{\partial u}{\partial y} = ik_y u, \qquad \frac{\partial^2 u}{\partial y^2} = -k_y^2 u.$$

Substituting into the wave equation yields

$$-\omega^2 u = -c^2(k_x^2 + k_y^2)u.$$

Since the exponential is nonzero, this holds if and only if

$$\omega^2 = c^2(k_x^2 + k_y^2).$$

For positive wave speed $c$, we choose the nonnegative frequency

$$\boxed{\omega = c\sqrt{k_x^2 + k_y^2}}.$$

Using $k_x = m_x\pi$ and $k_y = m_y\pi$, this becomes

$$\boxed{\omega = c\pi\sqrt{m_x^2 + m_y^2}}.$$

Thus, with this dispersion coefficient, the proposed wave satisfies the wave equation.